

How to build a plot

Dr. Margriet Groen

Overview

- What is a Quarto document and how do we use it?
- How to run code to read in data and build a plot

Step 2: Setup - load libraries

```
1 library(here)
2 library(tidyverse)
```

1. The `here` package helps with finding your files.
2. The `tidyverse` is a set of packages that handles almost everything you need to do with data.

Learn more about the `tidyverse` here: <https://tidyverse.org>

Step 3: Reading in the data

```
1 data_mw <- read_csv(here("PSYC4141/files/week1/", "data_bmw3.csv"))
```

1. `read_csv` is function that reads in data that are stored in a file of the type `.csv`. We need to tell R where the file is and what it is called.

`<-` is an assignment arrow, telling R that we want to put the data in an object (or dataframe) called `data_mw`

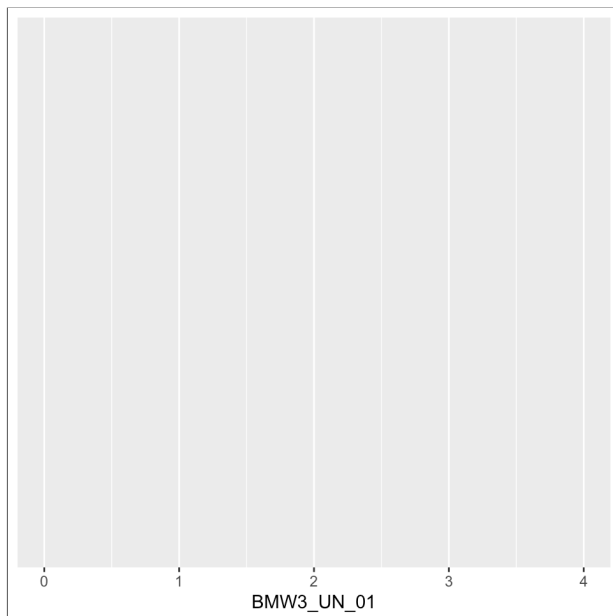
Step 4: Build the plot

Code

Plot

```
1 ggplot(data = data_mw, mapping = aes(x = BMW3_UN_01))
```

1. We use `ggplot()` to tell R that we want to create a plot and we tell R the data the use and the variable to plot



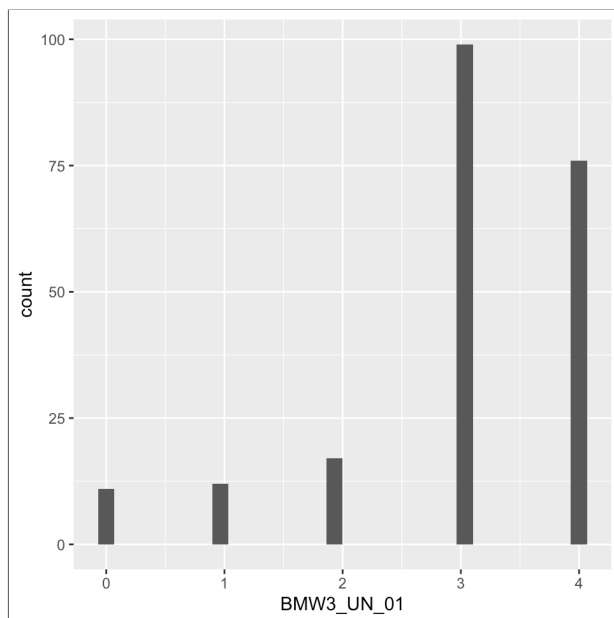
Step 4: a simple histogram

Code

Plot

```
1 ggplot(data = data_mw, mapping = aes(x = BMW3_UN_01)) +  
2   geom_histogram()
```

2. We tell R what type of plot to use



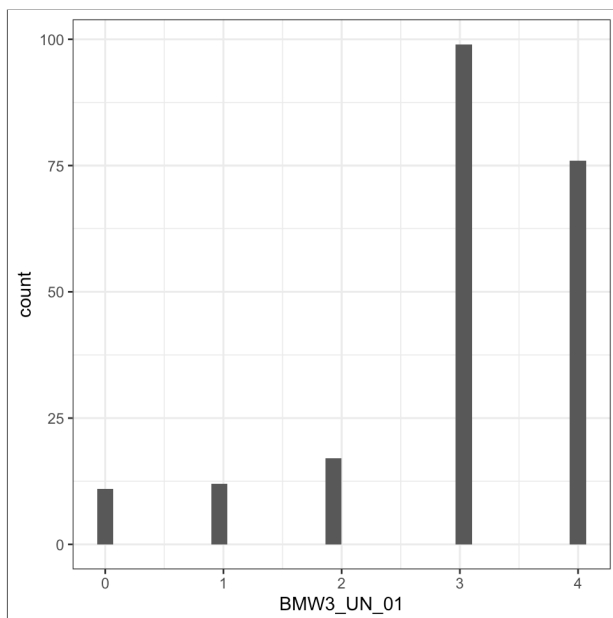
Step 4: Build the plot - different background

Code

Plot

```
1 ggplot(data = data_mw, mapping = aes(x = BMW3_UN_01)) +  
2   geom_histogram() +  
3   theme_bw()
```

3. Changes the background



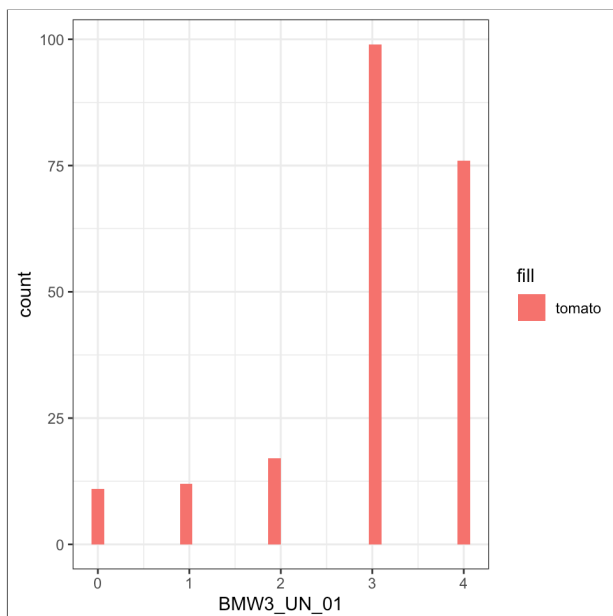
Step 4: Build the plot - change the colours of the bars

Code

Plot

```
1 ggplot(data = data_mw, mapping = aes(x = BMW3_UN_01, fill = "tomato")
2   geom_histogram() +
3   theme_bw()
```

1. Specify an additional bit of code (or **argument**) inside the **mapping** part of the function **ggplot()**. Instead of using the default grey, I'm telling R to 'fill in' the bars with a tomato red.



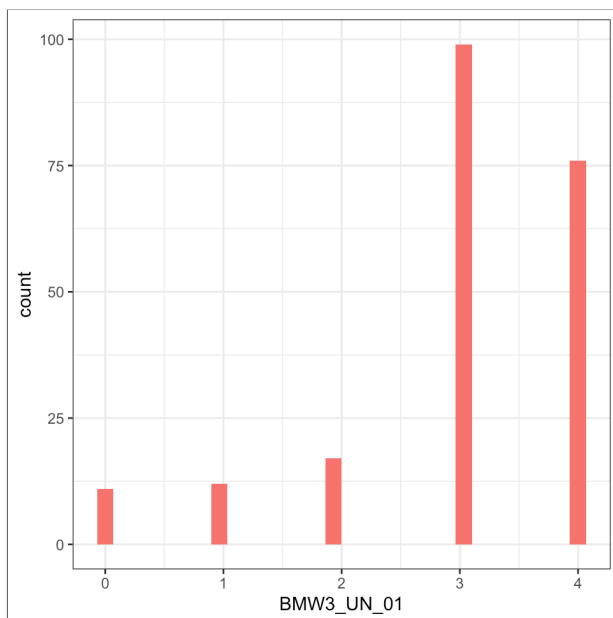
Step 4: Build the plot - remove the legend

Code

Plot

```
1 ggplot(data = data_mw, mapping = aes(x = BMW3_UN_01, fill = "tomato"))
2   geom_histogram() +
3   theme_bw() +
4   theme(legend.position = "none")
```

4. Removes the legend



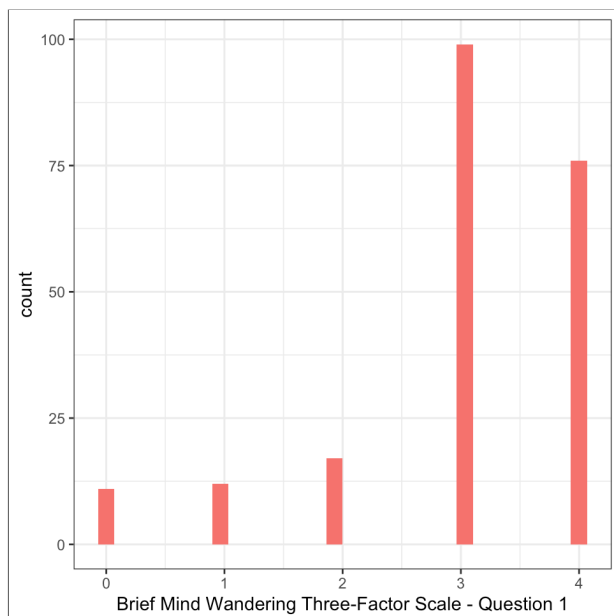
Step 4: Build the plot - change the title of the x-axis

Code

Plot

```
1 ggplot(data = data_mw, mapping = aes(x = BMW3_UN_01, fill = "tomato"))
2   geom_histogram() +
3   theme_bw() +
4   theme(legend.position = "none") +
5   labs(x = "Brief Mind Wandering Three-Factor Scale - Question 1")
```

5. Adds a more informative title for the x-axis.



Step 4: Build the plot - change the value labels

Code

Plot

```
1 ggplot(data = data_mw, mapping = aes(x = BMW3_UN_01, fill = "tomato"))
2   geom_histogram() +
3   theme_bw() +
4   theme(legend.position = "none") +
5   labs(x = "Brief Mind Wandering Three-Factor Scale - Question 1") +
6   scale_x_continuous(
7     breaks = 0:4,
8     labels = c("Fully disagree", "Somewhat disagree", "Neutral", "So
9   )
```

Lines 6. to 9. Replaces numerical values with text labels

